Perfect squares

1 State whether the following numbers are perfect squares:

a 6 **b** 25 **c** 49 **d** 44

e 18 **f** 144 **g** 36 **h** 12

2 Evaluate the following:

a 4^2 **b** 7^2 **c** 9^2 **d** 12^2

e $9^2 + 6$ **f** $3^2 + 7^2$ **g** $7^2 + 9^2$ **h** $100^2 + 10^2$

i $7^2 - 3^2$ **j** $9^2 - 8^2$ **k** $12^2 - 6^2$ **l** $100^2 - 10^2$

3 Harry was working out $2^2 \times 5^2$ and thought that he could simplify the expression using the fact that $2 \times 5 = 10$.

a Evaluate $2^2 \times 5^2$, by first expanding and evaluating each square.

b Now, using the fact $2 \times 5 = 10$, evaluate 10^2.

c Is $2^2 \times 5^2 = (2 \times 5)^2$ a true statement?

d Calculate $4^2 \times 2^2$ using the above method.

Square roots

4 Given that we know $5 \times 5 = 25$. What is $\sqrt{25}$?

5 Evaluate the following:

a $\sqrt{4}$ **b** $\sqrt{16}$ **c** $\sqrt{121}$ **d** $\sqrt{144} + \sqrt{64}$

e $\sqrt{144} - \sqrt{64}$ **f** $\sqrt{6^2 + 8^2}$ **g** $\sqrt{13^2 - 12^2}$ **h** $\sqrt{13^2} - \sqrt{12^2}$

Higher roots

6 Are the following statements true or false?

a $\sqrt[3]{1000} = 10$ **b** $\sqrt[3]{64} = 4$

7 Evaluate the following:

a $\sqrt[3]{27}$ **b** $\sqrt[3]{64}$ **c** $\sqrt[3]{1000}$ **d** $\sqrt[3]{343}$

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8 Evaluate the following:

a $\sqrt[4]{16}$ **b** $\sqrt[4]{256}$ **c** $\sqrt[5]{100\,000}$ **d** $\sqrt[5]{32}$

9 Solve $x^3 = 64$.

10 Find the side length of a cube with a volume of $27\ \text{cm}^3$.

11 Evaluate the following:

a $\sqrt[3]{512} \times \sqrt[3]{64}$ **b** $\sqrt[3]{64} \div \sqrt[3]{8}$

c $\sqrt[3]{8} \times \sqrt[3]{1000}$ **d** $\sqrt[3]{64} \div \sqrt{16}$

e $6^2 \div \sqrt[3]{8}$ **f** $\sqrt[3]{1000} \div \sqrt{100}$

g $8^3 \div \sqrt[3]{64}$ **h** $\sqrt[3]{125} \times \sqrt[5]{32} \div \sqrt[3]{8}$

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